Social Workplace AI

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Problem Statement

My research approaches algorithmic systems in the workplace by designing alternative interactions that augment human capabilities, labor practices and social relations while surfacing underlying mechanisms of control and oppression. From ride-hail drivers to artists, I design human-centered interactive systems that

(a) restore core worker values of security, fulfillment and dignity
(b) recognize and replace hidden means of engagement with worker-aligned incentives
(c) reconfigure asymmetric power relations in multi-stakeholder interactions

By invisibilizing worker labor and exploiting consumer attention, current forms of opaque workplace AI undermine our agency, cognitive capabilities and trust for one another. I build tools that help workers regain autonomy, connect communities of social support, and create counterspaces of resistance that force black-box algorithms and their surrounding ecosystem to acknowledge the human users who ultimately power their sustained existence.

Rapidly advancing AI capabilities are reshaping how we consume media, understand ourselves and interact with one another. For instance, Uber received seven cases of reported driver suicides in 2018 alone [1]; the 16-year-old Adam Raine hung himself after consulting ChatGPT-4o for months [2]; despite Lua Health's rigorous attempt to detect mental health issues in workplace communications, workers quickly preferred more familiar, general-purpose chatbots [3]. While these cautionary tales seem daunting – especially given the introduction of ride-hail more than a decade ago – I envision future mitigating interventions that better prioritize worker dignity, incorporate diverse stakeholder preferences and cultivate communities of social support. To prevent disruptive technologies from eroding our cognitive human capabilities and social relations, I propose to design **prosocial and multi-stakeholder interactive systems** that go beyond extractive practices to actively upskill, strengthen and prepare future workforces for an era of evolving advances. In addition to observational, critical and design methods, I plan to also implement functional systems that train and guide our collective autonomy toward more meaningful work, trusting social relations, and collaborative communities.

Preliminary research

My work identified that algorithmic systems pose a direct threat to human autonomy across contexts. Whether the user is a ride-hail driver, passenger or artist, platforms leverage and commodify their attention, data and labor to produce value — thereby eroding their physical, cognitive and creative agency. Drivers are held accountable for service quality by passenger ratings. At the same time, passengers also contribute critical location, demographic and preference data – which platforms leverage to escalate and enforce service expectations. Similarly, when artists search for inspiration or guidance, generative feedback also exploit and retrain on their styles and creative labor. As the digital creator economy expands, I plan to apply insights from the ride-hail context toward more creative professionals who work with more generative tools (*e.g.,* POET toward text-to-image generation), in pursuit of developing a unifying, humanistic framework that respects core human values, empowers negotiation and fosters supportive social relations as workers navigate the increasingly complex digital landscape.

In ridehail, my research demonstrated how creative interventions carry potential for transforming passenger perspectives and improving drivers' latent (and ultimately material) working conditions. Scholars from across the globe documented and critiqued the subpar working conditions and inadequate government regulations. In a series of 9 workshops with 19 passengers and 15 drivers, we designed six interactive in-ride games to probe how playful in-ride interventions can evoke joyful social interactions with passengers – motivating them to understand, engage with and advocate for the invisible labor, logistics and costs behind every ride. For passengers, these playful interactions surfaced knowledge gaps, prompted reflection and shifted perceptions around consumption behaviors. For drivers, these prototypes highlighted their preferences for more immersive and context-aware technologies. The emergent driver-passenger interactions also revealed opportunities to design more safe and appropriate ways of motivating prosocial and solidaristic behaviors. By analyzing interaction dynamics, we showed how platforms amplified the asymmetric power relations between the passenger and driver (via mechanisms like ratings), giving rise to increasing consumer expectations of service quality while stripping drivers of dignity and autonomy [4].

These findings carry implications for human-centered workplace technologies that generalize beyond service-based labor. As the digital creator economy expands, data creation and consumption are increasingly common and integrated practices across domains such as art, design and knowledge work more broadly. In these professions, it becomes even more critical to engage more in-depth and critical metacognitive human capabilities, which are important for planning, monitoring and making decisions around work. I aim to develop creative and socially approachable interventions that augment such cognitive human capabilities to engage the creative and critical reflection of individual workers, as well as more prosocial practices that build collective support and accountability among working communities.

Creative professionals (especially those producing visual content) represent another worker group whose labor outputs are increasingly monetized and appropriated by emerging generative technologies. To combat normative model biases, personalize outputs and increase user agency during early ideation stages of creation, I led a team of interdisciplinary researchers across HCI and AI to design and evaluation of POET, an interactive tool that (1) automatically discovers visual homogeneities in text-to-image generative models, (2) expands these to diversify generated output images, and (3) learns from user feedback to personalize expansions [5]. By juxtaposing a single image output with more visually diverse alternatives, our cross-domain user study with 28 participants showed how POET offered creators alternative choices during idea generation, prompting more deliberate and reflective decision-making while efficiently supporting more pluralistic and personalized values. This combination of slow and fast thinking mirrors how metacognition mechanisms improve reasoning of model architectures [6].

Proposed Methods

To scale up this framework, I propose to develop lightweight interactions that promote more in-depth reflection among multiple stakeholder groups. In particular, I envision eliciting and incorporating diverse user preferences at *individual*, *relational* and *community-based* levels to foster more intentional, context-aware and deliberative decision-making as workers adapt to advancing (generative) model capabilities.

Designing Interactions For Autonomy & Upskilling Individual Capabilities. Both algorithmic and generative systems are taking increasingly ubiquitous roles in both crowdbased and complex forms of work. To prepare future workforces, I aim to apply theories from education and developmental psychology to develop interventions that actively train workers to acquire and practice (meta)cognitive skills that support their better planning, monitoring and decision making when working with such systems. In particular, I plan to borrow techniques from fields such as behavioral economics and mechanism design to develop incentives that align with core worker objectives of purpose, dignity and financial security. For low-wage and algorithmically-managed work, the ability to reflect on and resist low quality work is an especially important skill. The presence of alternatives helped POET users achieve better outcomes, and similar approaches can teach workers to refuse poor quality work.

Motivating Collectivism & Accountability with Lightweight Prosocial Interactions. To effectively engage worker upskilling, they must first be *motivated* to self-improve. One way to help workers achieve behavioral change is to enact social accountability from stakeholders who are beneficiaries of their services (*e.g.,* consumers, platform shareholders, governments). Gamification was one approach that showed promise for facilitating consumer behavioral change. I plan to scale up this intervention to create more joyful, reflective and prosocial interactions with stakeholders to motivate solidaristic actions that challenge worker oppression. For instance, how can we leverage mixed/virtual realities to simulate individual worker narratives? How can digital tools, interactions and spaces promote more responsible technology use and consumption while allowing citizens to enforce corporate social responsibilities from the ground up?

One key challenge currently preventing workers from more information-sharing practices is the high effort required to document invisible labor. Emergent modes of lightweight and ephemeral interactions (*e.g.* voice/video-based chatrooms that capture paralinguistic visual cues) offer promise for achieving (voice) activism asynchronously. How can such alternative modes, including emerging video- and world-generation capabilities, capture or re-enact labor accountabilities? With such multi-stakeholder discourse data, I plan to apply analysis techniques (*e.g.* power, conversation) and frameworks (*e.g.* postcolonial) from fields such as organizational management and critical computing to analyze the complex power relations at work.

Building Connected Communities of Social Support. The disembedded nature of platform-based work isolated workers from social support that is traditionally found among networks of coworkers, resulting in atomized careers that harm psychological well-being over time. To build communities of trust, we must retrain workers to leverage digital communication tools in ways that facilitate relational support, instead of only messaging for work purposes. As work arrangements become increasingly flexible and remote, the affective ties between workers and organizations are gradually waning. How can we redesign both digitally-mediated and in-person social interactions within modern, reconfigured work arrangements to counteract isolation and advance labor rights? How can agent-based simulations model and promote more deliberative and pluralistic practices when building protective and regulatory policies? How can 3D digital environments simulate environments of community support? Moving forward, I am eager to develop open-sourced systems that prototype such community-based and prosocial civil engagements.

Citations

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